Query Parameter Models

If you have a group of **query parameters** that are related, you can create a **Pydantic model** to declare them.

This would allow you to **re-use the model** in **multiple places** and also to declare validations and metadata for all the parameters at once. 😎

> **Note**
>
> This is supported since FastAPI version `0.115.0`. 🤓

Query Parameters with a Pydantic Model

Declare the **query parameters** that you need in a **Pydantic model**, and then declare the parameter as `Query`:

Python 3.10+

```python
from typing import Annotated, Literal

from fastapi import FastAPI, Query
from pydantic import BaseModel, Field

app = FastAPI()


class FilterParams(BaseModel):
    limit: int = Field(100, gt=0, le=100)
    offset: int = Field(0, ge=0)
    order_by: Literal["created_at", "updated_at"] = "created_at"
    tags: list[str] = []


@app.get("/items/")
async def read_items(filter_query: Annotated[FilterParams, Query()]):
    return filter_query
```

Ask AI

Other versions and variants

Python 3.10+ - non-Annotated

Tip

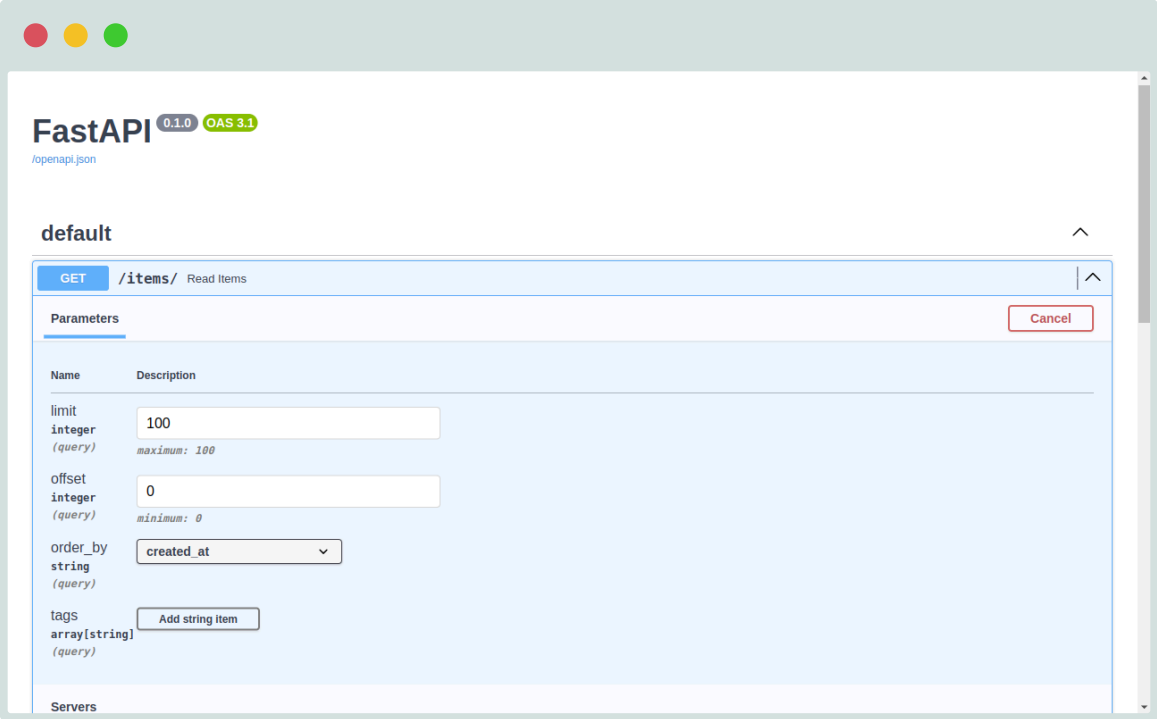
Prefer to use the `Annotated` version if possible.

```
from typing import Literal

from fastapi import FastAPI, Query
from pydantic import BaseModel, Field

app = FastAPI()


class FilterParams(BaseModel):
    limit: int = Field(100, gt=0, le=100)
    offset: int = Field(0, ge=0)
    order_by: Literal["created_at", "updated_at"] = "created_at"
    tags: list[str] = []


@app.get("/items/")
async def read_items(filter_query: FilterParams = Query()):
    return filter_query
```

FastAPI will **extract** the data for **each field** from the **query parameters** in the request and give you the Pydantic model you defined.

Check the Docs

You can see the query parameters in the docs UI at `/docs`:



Forbid Extra Query Parameters

In some special use cases (probably not very common), you might want to **restrict** the query parameters that you want to receive.

You can use Pydantic's model configuration to `forbid` any `extra` fields:

Python 3.10+

```python
from typing import Annotated, Literal

from fastapi import FastAPI, Query
from pydantic import BaseModel, Field

app = FastAPI()


class FilterParams(BaseModel):
    model_config = {"extra": "forbid"}

    limit: int = Field(100, gt=0, le=100)
    offset: int = Field(0, ge=0)
    order_by: Literal["created_at", "updated_at"] = "created_at"
    tags: list[str] = []


@app.get("/items/")
async def read_items(filter_query: Annotated[FilterParams, Query()]):
    return filter_query
```

🤓 **Other versions and variants**

Python 3.10+ - non-Annotated

Tip

Prefer to use the `Annotated` version if possible.

```python
from typing import Literal

from fastapi import FastAPI, Query
from pydantic import BaseModel, Field

app = FastAPI()


class FilterParams(BaseModel):
    model_config = {"extra": "forbid"}

    limit: int = Field(100, gt=0, le=100)
    offset: int = Field(0, ge=0)
    order_by: Literal["created_at", "updated_at"] = "created_at"
    tags: list[str] = []


@app.get("/items/")
async def read_items(filter_query: FilterParams = Query()):
    return filter_query
```

If a client tries to send some **extra** data in the **query parameters**, they will receive an **error** response.

For example, if the client tries to send a `tool` query parameter with a value of `plumbus`, like:

```
https://example.com/items/?limit=10&tool=plumbus
```

They will receive an **error** response telling them that the query parameter `tool` is not allowed:

```
{
    "detail": [
        {
            "type": "extra_forbidden",
            "loc": ["query", "tool"],
            "msg": "Extra inputs are not permitted",
            "input": "plumbus"
        }
    ]
}
```

Summary

You can use **Pydantic models** to declare **query parameters** in **FastAPI**. 😎

> **Tip**
>
> Spoiler alert: you can also use Pydantic models to declare cookies and headers, but you will read about that later in the tutorial. 🤫